AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 2 · TEACHER GUIDE

Dot products and weighted sums

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Make every contribution visible.

Use complete matched-product traces and validated-input contracts;distinguish score arithmetic from learned or probabilistic claims.

Keep one habit: Assess signed products,total,controlled change and dimension guard with the checked independent key.

MAIN LESSON · SECTION 1

Prepare the connected mathematical investigation

> Keep explanation,shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Calculate a **dot product** by matching feature positions

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

Their components assign different values to width and height. A calculation using different **weight**s for those features can produce different results.

Opening retrieval

Without opening the old lesson, explain one key idea from Representing **input**s as vectors. Give an example and a mistake that the idea helps you avoid.

Look for: Encode a record using a fixed feature order and units

Reference example, not the only acceptable wording: Teacher explanation: A [4, 2], B [2, 4], C [3, 3]. Each has dimension 2. The table has 3 rows and 2 columns.

Pictured end goal

By the end, I can explain **dot product**s and **weight**ed sums using a traced example and a stated limitation.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Teach the learning cycle

Retrieval answer guidance

Representing inputs as vectors

Teach the **dot product** as a traceable sum of signed **contribution**s in two 45-minute sessions. Keep **weight**s fixed throughout each comparison and separate hand-chosen parameters from **training**.

Students should explain a **score** change using the corresponding coefficient, while retaining the limits of that interpretation.

Outcomes

Calculate a **dot product** by matching feature positions

Explain how a **weight** changes a **score** with other **input**s fixed

Distinguish a **weight**ed **score** from a probability or trained model
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Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach matched products,compatible units and a bounded score

> Keep the parcel and brightness coefficient sets explicit.

Main explanation,full trace and guided check

A small model can combine several measurements into one **score**. Let x = [4, 2] contain fictional width and height measurements in cm. Set **weight**s w = [2, -1].

Multiply matching positions, then add: s = (2 times 4) + (-1 times 2) = 6. The letter x names the **input** vector.

We call this calculation a **dot product**, written w dot x. The vectors must have the same number of components. Each **weight** belongs to one named feature.

These hand-selected **weight**s demonstrate a computation; no **training** has happened. The result 6 is a **score**, not a probability or a claim that this parcel is safe.

The coefficients carry **score**-per-cm units in this example, making both **contribution**s compatible **score** units. A **weight**ed sum needs a consistent definition of units.

Keep w = [2, -1] fixed. For A = [4, 2], the **score** is 6. For B = [2, 4], it is 4 - 4 = 0. For C = [3, 3], it is 6 - 3 = 3. Swapping **input** positions without swapping their matching **weight**s changes the computation.

A zero **score** can come from cancellation, even when neither **input** is zero.

The length check matters: Python zip alone stops at the shorter list and could silently omit a feature. The function assumes numeric **input**s already validated under the schema.

First predict the three printed values on paper, then compare.

Parcel input [width_cm, height_cm]	Width contribution 2×width	Height contribution−1×height	Total score
A[4, 2]	8	-2	6
B[2, 4]	4	-4	0
C[3, 3]	6	-3	3
Same-width changed height[4, 3]	8	-3	5

Multiply matching components then add

POSITION	CONTRIBUTION RULE	WORKED CONTRIBUTION
1 width	2 times width	2 times 4 gives 8
2 height	-1 times height	-1 times 2 gives -2
Add both	8 plus -2	Score 6

The first two rows are separate contributions. The last row combines them; multiplying the final input sum by the final weight sum would be a different operation.

Guided: [4, 3] gives 8 and -3, total 5, down 1 from 6 because height rose by 1 with coefficient -1. Exit: the **score** has no supplied probability mapping, fitted probability model or calibration **evidence**.

Multiplying it by ten does not create a supported probability.

Separate pictured brightness facts

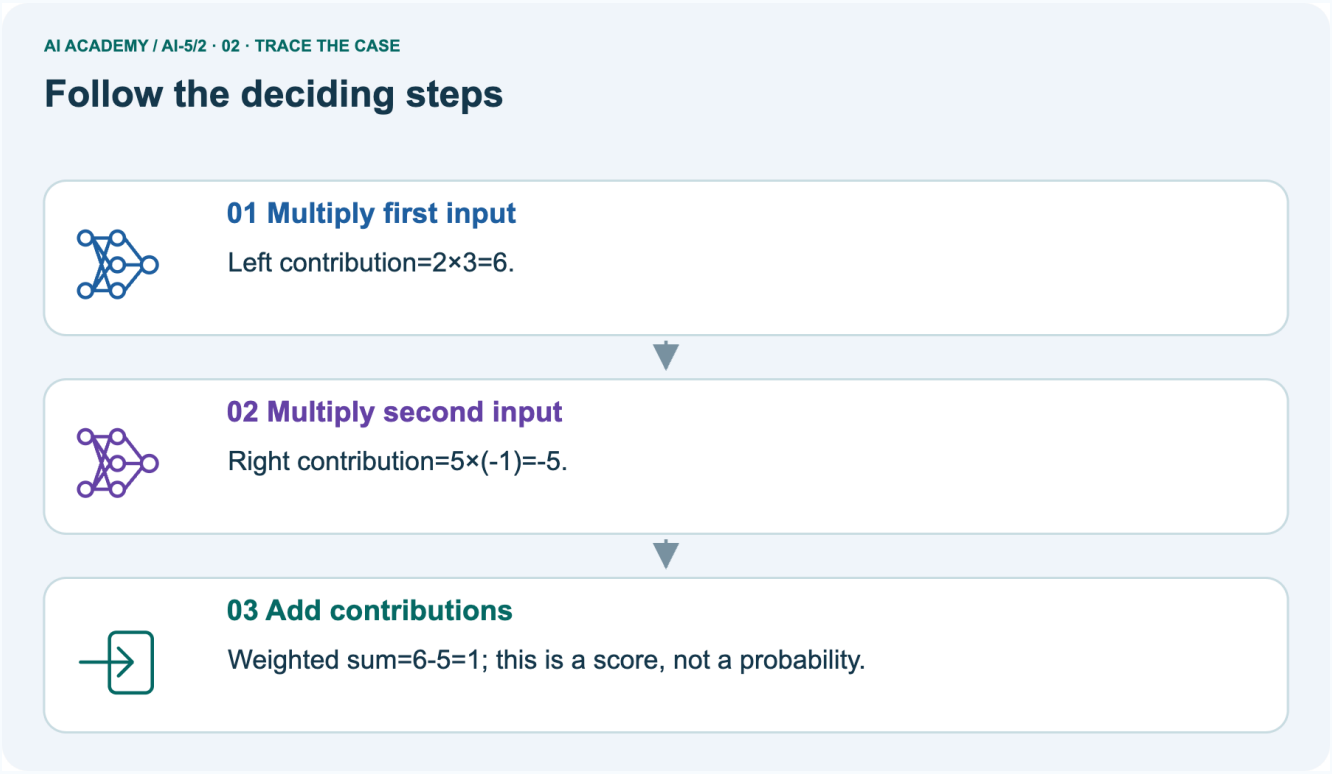
Draw each **input**'s **contribution** to a toy **weight**ed brightness **score**.

Inputs [2, 5]; **weight**s [3,-1]; no bias. **Score** is 2×3+5×(-1).

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

This brightness picture uses x=[2, 5], w=[3, −1], no bias:contributions 6 and−5, total 1. Increasing only the right input to6 changes that contribution to−6, total 0. The input still has nonzero measurements; the contributions cancel. Main parcel examples instead use w=[2, −1], so do not swap the two weight lists. A score can be negative, zero or positive; no probability interpretation or training is supplied.

Trace negative contribution and changed score



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Multiply first input	Left contribution=2×3=6.
2. Multiply second input	Right contribution=5×(-1)=-5.
3. Add contributions	Weighted sum=6-5=1; this is a score, not a probability.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A **weight** controls how strongly an **input** contributes to the **score**. The left connection adds three times its measurement; the negative right connection subtracts its measurement. Calculate each **contribution** before adding them.

This **dot product** is a signed **score**; it becomes a probability only if a separate, justified transformation and interpretation are supplied.

These declared **weight**s are hand-chosen toy parameters; no network **training** or calibrated probability is claimed.

Change: Increase only the right **input** from 5 to6.

Prediction: The **score** becomes 6-6=0.

Reason: The right **weight** is-1, so adding 1 to that **input** lowers the **score** by1.

▼ Earlier wording, alternatives and picture transcripts

Term	Meaning
Weight	A numerical coefficient multiplying one input component
Dot product	The sum of products of corresponding vector components
Contribution	The product of one input component and its matching weight
Score	A numerical output whose interpretation must be specified

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Multiply matching components then add

POSITION	CONTRIBUTION RULE	WORKED CONTRIBUTION
1 width	2 times width	2 times 4 gives 8
2 height	-1 times height	-1 times 2 gives -2
Add both	8 plus -2	Score 6

The first two rows are separate contributions. The last row combines them; multiplying the final input sum by the final weight sum would be a different operation.

> **Keep w = [2, -1]. Change [4, 2] to [4, 3]. Compute both contributions and explain the score change.**

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Dot products and weighted sums

The facts we will use

AI ACADEMY / AI-5/2

Weighted brightness score

Input 2 x weight 3 = 6

Input 5 x weight -1 = -5

Add the products and bias 0: total = 1

Keep the total (identity) -> output 1

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/2

Dot products and weighted sums

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Multiply first input

Left contribution=2 x 3=6.

2. Multiply second input

Right contribution=5 x (-1)=-5.

3. Add contributions

Weighted sum=6-5=1; this is a score, not a probability.

CHANGE ONE THING

Increase only the right input from 5 to6.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: For w = [2, -1] and x = [2, 4], explain why **score** zero does not mean the **input** contains no information.

Teacher check: Teacher explanation: **Contribution**s 4 and -4 cancel. The **input** still contains width 2 and height 4; zero is the combined **score**.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Dot product**s and **weight**ed sums. Proposed explanation: “Multiply the sums instead”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: The operation is 2 times 4 plus -1 times 2, not (2 - 1) times (4 + 2). That shortcut happens to match for some **input**s but is not the rule.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Which connection explains why more right brightness lowers this **score**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

The right connection has negative **weight** -1. Its **contribution** changes from -5 to-6 while the left **contribution** stays 6. Changed-case check: The **score** becomes 6-6=0.

The right **weight** is-1, so adding 1 to that **input** lowers the **score** by1.

Changed-case reasoning: The **score** becomes 6-6=0. The right **weight** is-1, so adding 1 to that **input** lowers the **score** by1.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide three labelled **contribution** boxes and a number line for signed addition.

Extension: permute both **input**s and matching **weight**s together and show that the sum is preserved; then permute only one to expose a schema mismatch.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1 start to 7 minutes	Ask the prerequisite question. Check signed multiplication with -1 times 2 before introducing a dot product.
7 to 18 minutes	Lay out the contribution table. Students draw a connection between each named input and its own weight.
18 to 32 minutes	Complete W1 and W2. Trace A, B and C with the same weights; discuss cancellation rather than calling zero an empty input.
32 to 45 minutes	Run or trace the guarded Python function. Try a missing weight and explain the error. Complete the guided change.
Session 2 start to 8 minutes	Retrieve the two contributions for A. Ask what changes if height increases while width stays fixed.
8 to 20 minutes	Complete W3, converting only height to mm. Show why weight magnitude depends on units.
20 to 35 minutes	Collect W4 independently. Require the three signed contributions, original and changed totals, and the dimension error explanation.
35 to 45 minutes	Complete W5 and the exit response. Use the fresh retry after targeted feedback; keep first attempts for assessment.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Normal1, cancellation 0, zero coefficient 6, mismatched dimensionsValueError; missing input violates the numeric contract. Empty lists yield 0 arithmetically but are not the two-feature task. Unit-preserving repair weights[2, –.1], input[4, 20]score 6. Independent products–6, +4, +1 total–1; second input 1→2 total 3, change+4; three features require three weights. Do not equate score 6 with60%probability.

Diagnose the current misconceptions

Multiply the sums instead: The operation is 2 times 4 plus -1 times 2, not (2 - 1) times (4 + 2). That shortcut happens to match for some **input**s but is not the rule.

Negative means unimportant: A negative coefficient changes direction. Units, data and the rest of the model affect interpretation.

The **weight**s have been learned: These values were supplied. **Training** would require data, an objective and a **fitting** procedure.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows	Next teaching action
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new schema or calculation transfer

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Use x = [3, 1, 2] and w = [-2, 4, 0.5]. Give each **contribution** and the total. Increase only the second component from 1 to 2 and compute the new total. Explain why a two-**weight** list is invalid for this **input**.

Teacher explanation: **Contribution**s -6, 4, 1 give -1. With second component 2, **contribution**s -6, 8, 1 give 3, a rise of 4.

Three **input** components need three matching **weight**s; silently dropping the final component changes the computation.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Calculate contributions

For w = [2, -1] and x = [3, 5], write both **contribution**s and their sum.

Teacher explanation: **Contribution**s are 6 and -5, so the **score** is 1.

W2 Explain cancellation

For w = [2, -1] and x = [2, 4], explain why **score** zero does not mean the **input** contains no information.

Teacher explanation: **Contribution**s 4 and -4 cancel. The **input** still contains width 2 and height 4; zero is the combined **score**.

W3 Preserve a unit change

Width stays in cm and height changes to mm. For the worked parcel [4 cm, 20 mm], give the revised **weight**s and **score**.

Teacher explanation: Use [2, -0.1]. **Contribution**s 8 and -2 give **score** 6, preserving the original measurement meaning.

W5 Find lost information

Verify that [4, 2] and [5, 4] both **score** 6 with w = [2, -1]. Can a later step receiving only this **score** distinguish those two records?

Teacher explanation: The **score**s are 8 - 2 = 6 and 10 - 4 = 6. A deterministic later step receiving only 6 receives identical information and cannot distinguish them.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Guided and exit questions

Guided: [4, 3] gives 8 and -3, total 5, down 1 from 6 because height rose by 1 with coefficient -1. Exit: the **score** has no supplied probability mapping, fitted probability model or calibration **evidence**.

Multiplying it by ten does not create a supported probability.

W1 Calculate contributions

W2 Explain cancellation

W3 Preserve a unit change

W4 Compute a new dot product independently

Use x = [3, 1, 2] and w = [-2, 4, 0.5]. Give each **contribution** and the total. Increase only the second component from 1 to 2 and compute the new total. Explain why a two-**weight** list is invalid for this **input**.

Teacher explanation: **Contribution**s -6, 4, 1 give -1. With second component 2, **contribution**s -6, 8, 1 give 3, a rise of 4.

Three **input** components need three matching **weight**s; silently dropping the final component changes the computation.
</details>

W5 Find lost information

Independent assessment and fresh retry

W4 requires **contribution**s -6, 4, 1 and total -1, changed total 3 and a dimension check. Fresh retry: w = [1, -3, 2], x = [2, 1, 4] gives 2 - 3 + 8 = 7. If only the third **input** falls to 3, the **score** is 5.

Ask for the two-point decrease explanation.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect,support and revisit the mathematics

> Use the saved evidence to choose the next step.

Project and next-week bridge

Add a labelled **dot product**s and **weight**ed sums **evidence** sheet to Image-pattern Network Investigation. Show how you can calculate a **dot product** by matching feature positions.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide three labelled **contribution** boxes and a number line for signed addition.

Extension: permute both **input**s and matching **weight**s together and show that the sum is preserved; then permute only one to expose a schema mismatch.

Week 3 adds a bias after the **dot product** and uses an explicit decision boundary.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Delayed return and support record

Suggested return points in this level: ai-5/3, ai-5/5, ai-5/8. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Calculate a **dot product** by matching feature positions

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: For w = [2, -1] and x = [2, 4], explain why **score** zero does not mean the **input** contains no information.

Reasoned practice answer: Teacher explanation: **Contribution**s 4 and -4 cancel. The **input** still contains width 2 and height 4; zero is the combined **score**.

Check every learner before discussion: Use the worked case for **Dot product**s and **weight**ed sums. Proposed explanation: “Multiply the sums instead”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: The operation is 2 times 4 plus -1 times 2, not (2 - 1) times (4 + 2). That shortcut happens to match for some **input**s but is not the rule.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Increase only the right **input** from 5 to6. Require a **prediction** and a reason before comparing. Expected result: The **score** becomes 6-6=0. Reason: The right **weight** is-1, so adding 1 to that **input** lowers the **score** by1.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery **score**s. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

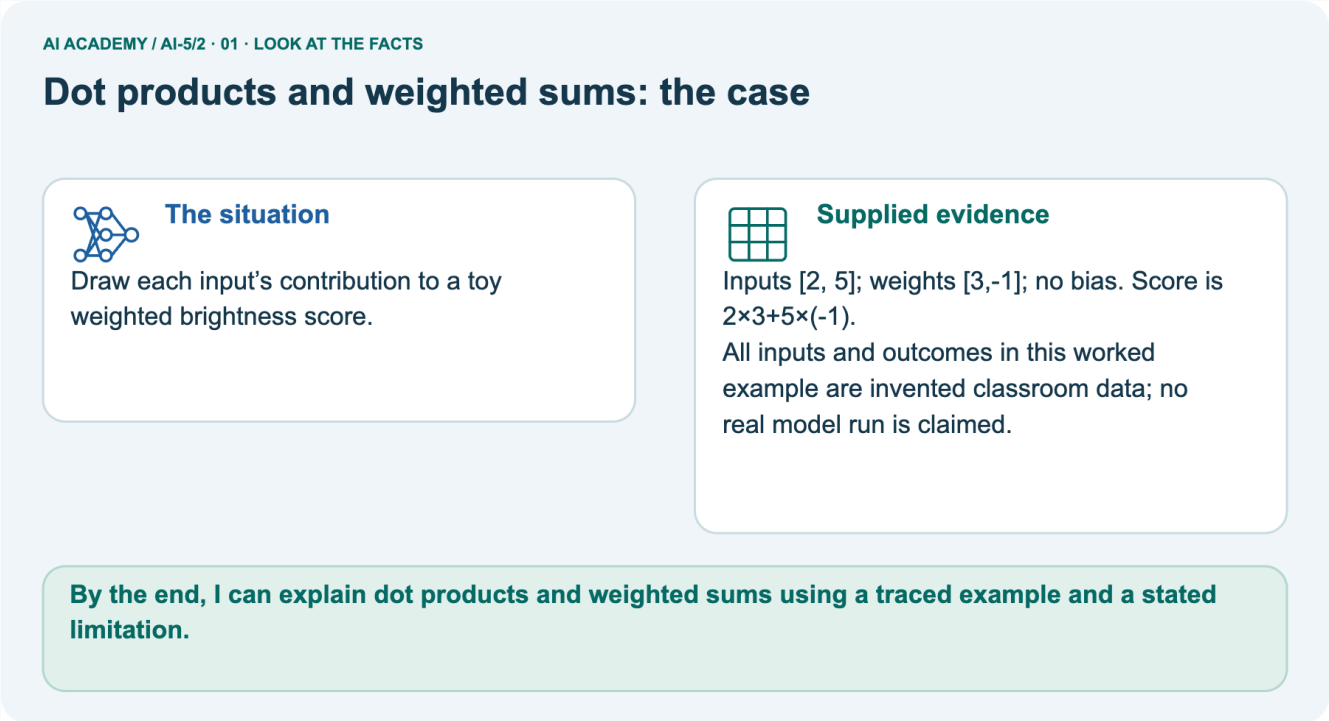
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Dot products and weighted sums: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

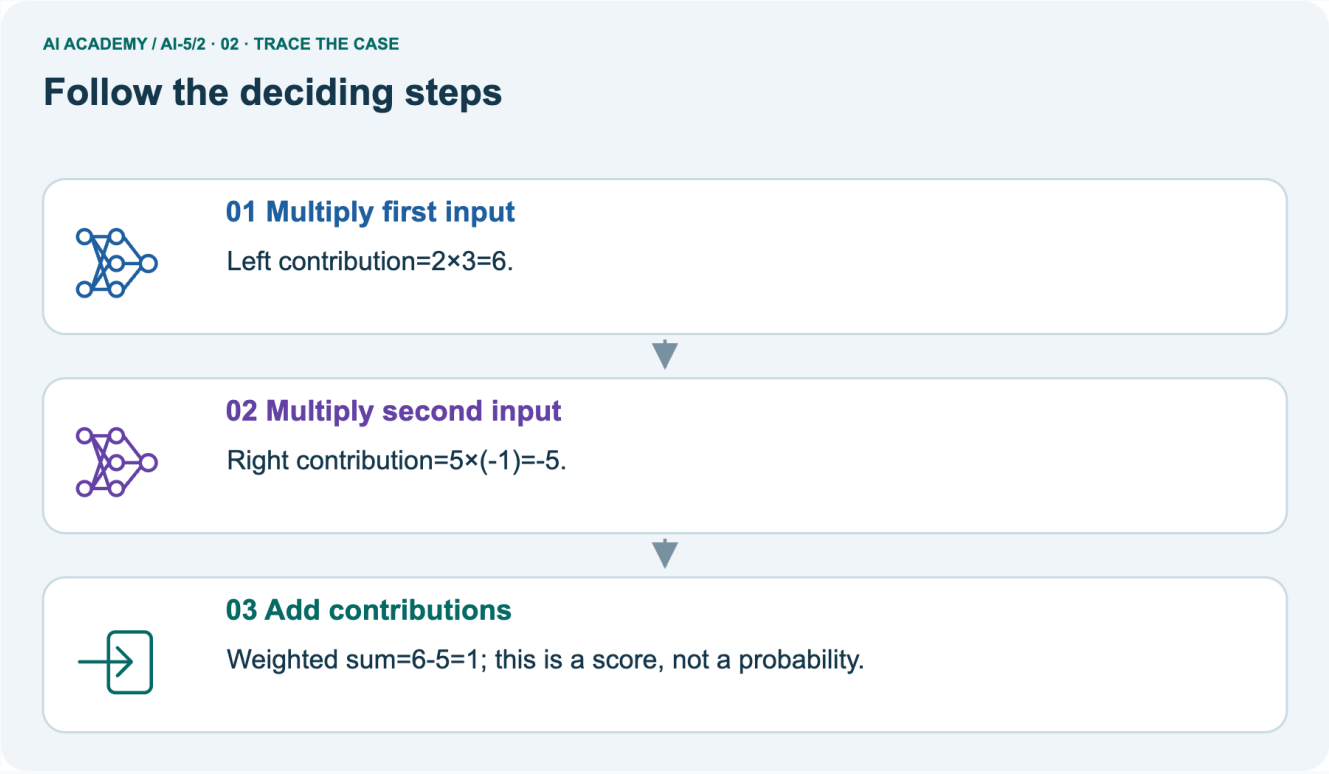
Draw each **input**'s **contribution** to a toy **weight**ed brightness **score**. Supplied facts: **Input**s [2, 5]; **weight**s [3,-1]; no bias. **Score** is 2×3+5×(-1).

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



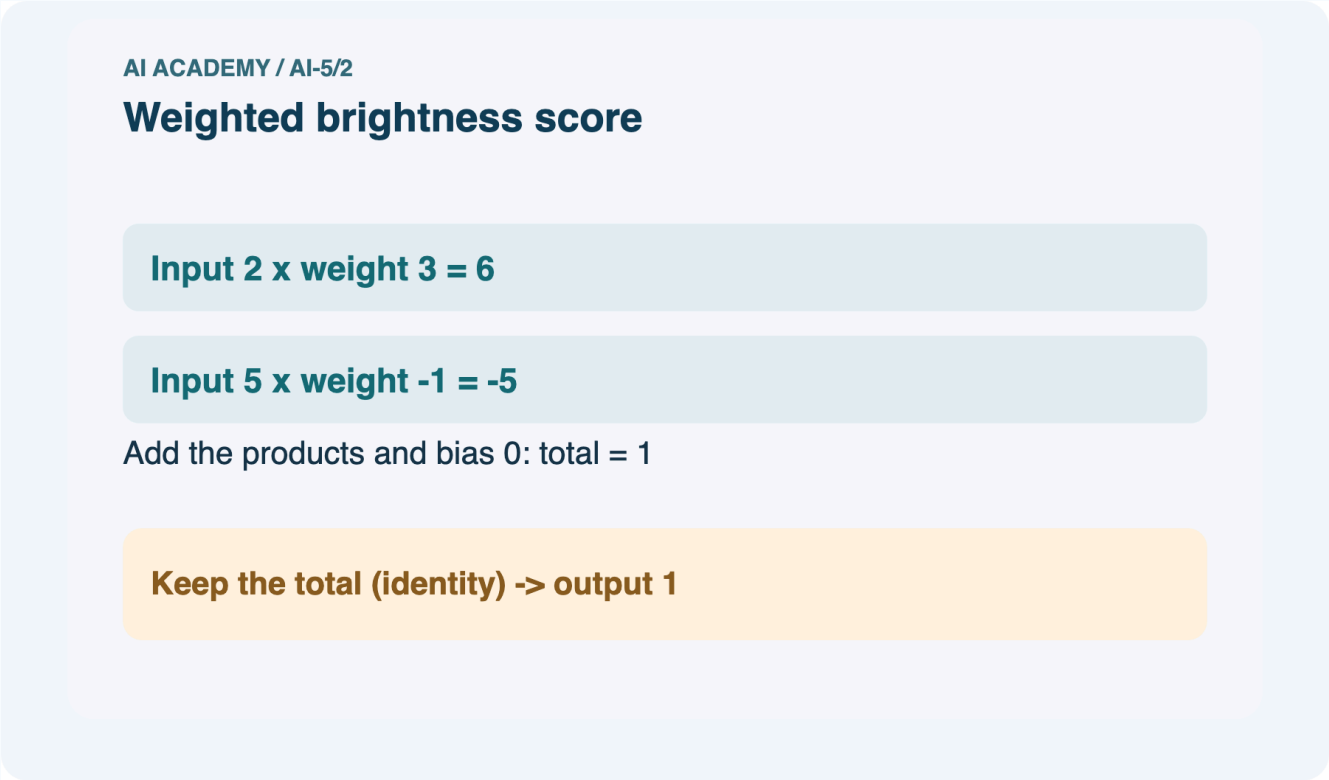
Follow the labelled picture with the matching explanation. Open the vector diagram

Multiply first **input**: Left **contribution**=2×3=6. → Multiply second **input**: Right **contribution**=5×(-1)=-5. → Add **contribution**s: **Weight**ed sum=6-5=1; this is a **score**, not a probability.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Weighted brightness score



Follow the labelled picture with the matching explanation. Open the vector diagram

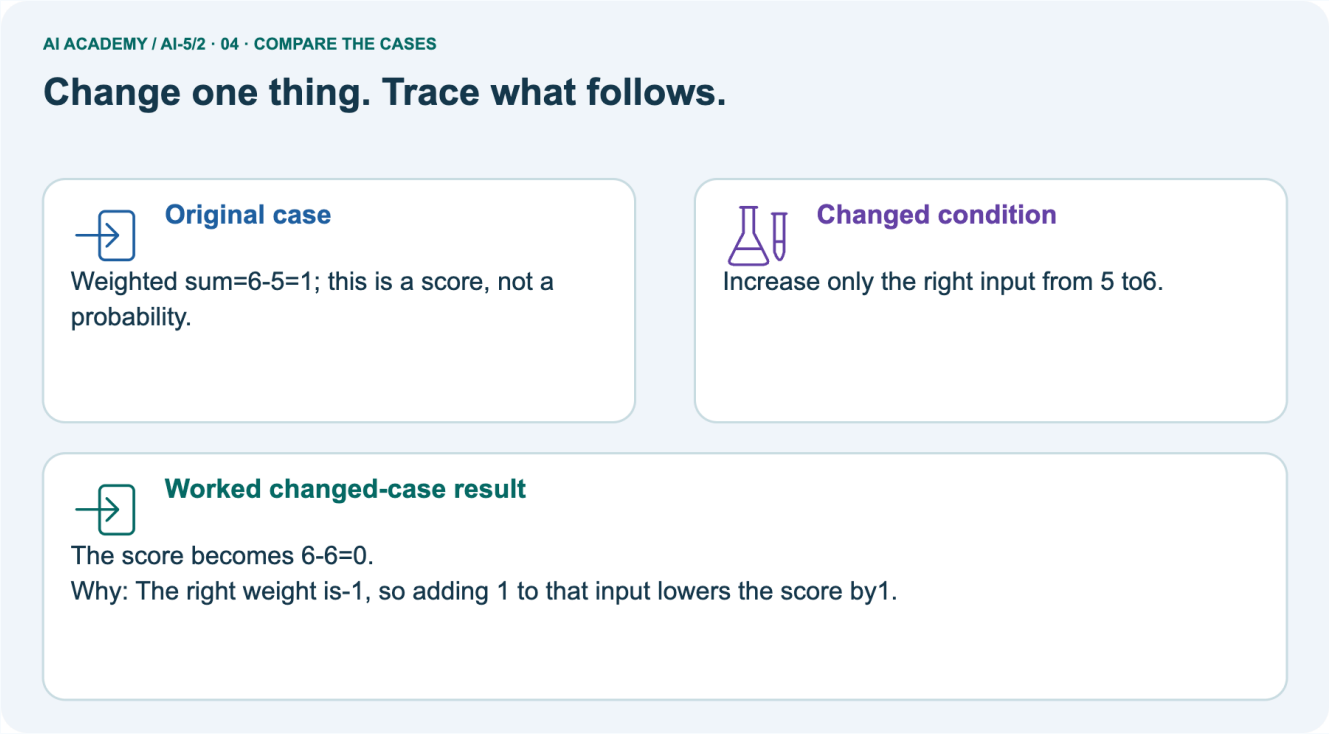
Weighted brightness **score**. **Input**s: [2, 5]; **weight**s: [3, -1]; bias: 0; activation: identity; **output**: 1.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: **Weight**ed sum=6-5=1; this is a **score**, not a probability. Changed condition: Increase only the right **input** from 5 to6. Expected result: The **score** becomes 6-6=0.

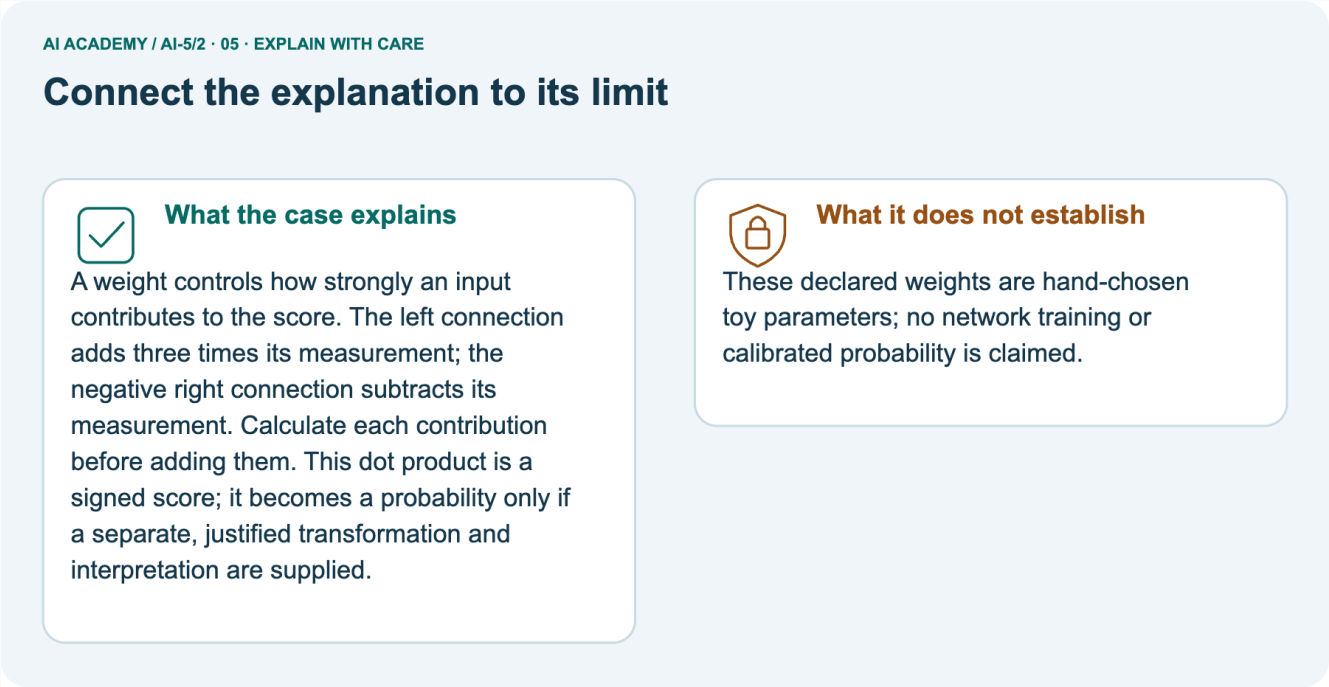
Reason: The right **weight** is-1, so adding 1 to that **input** lowers the **score** by1.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The right **weight** is-1, so adding 1 to that **input** lowers the **score** by1.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **weight** controls how strongly an **input** contributes to the **score**. The left connection adds three times its measurement; the negative right connection subtracts its measurement. Calculate each **contribution** before adding them.

This **dot product** is a signed **score**; it becomes a probability only if a separate, justified transformation and interpretation are supplied.

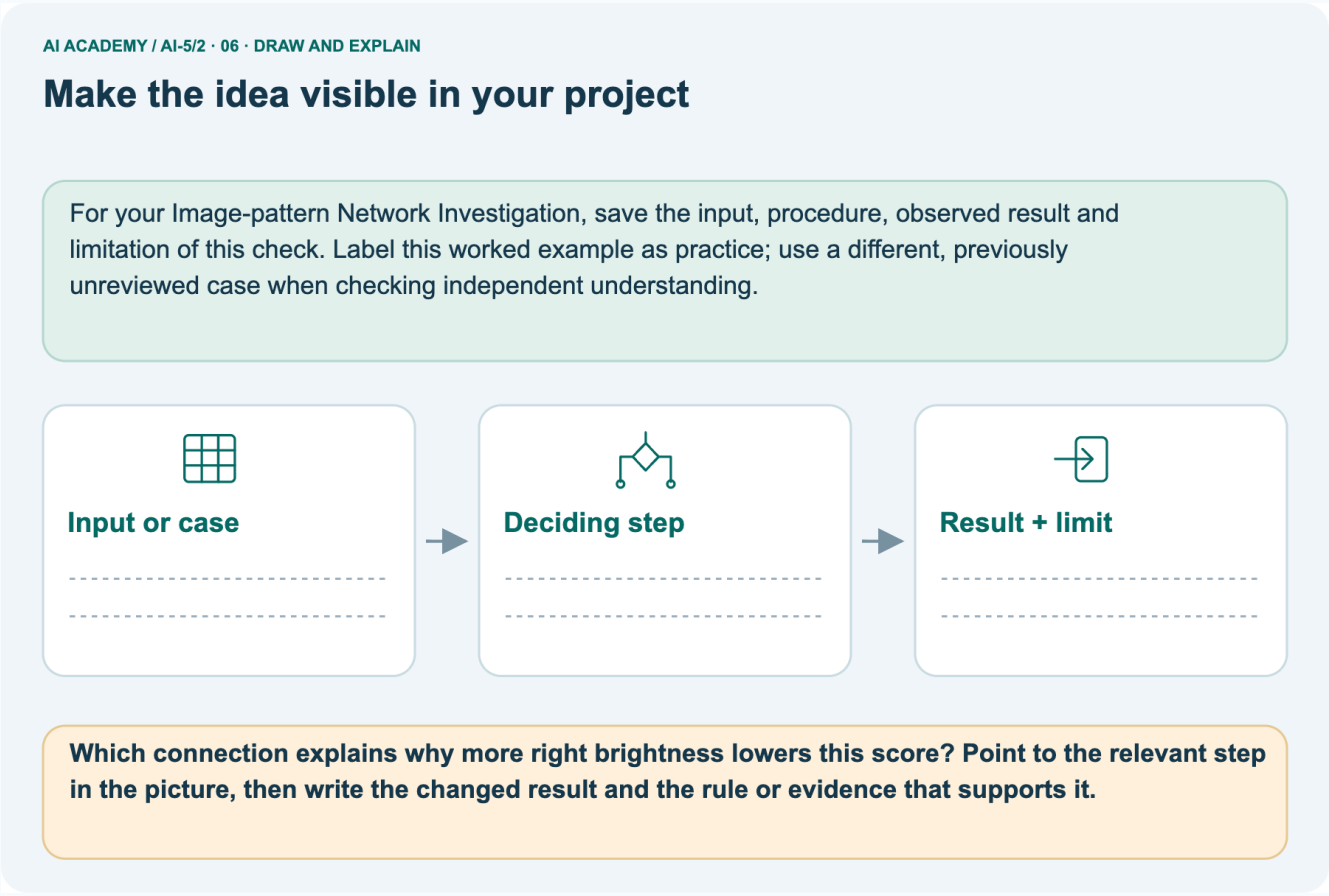
Limit: These declared **weight**s are hand-chosen toy parameters; no network **training** or calibrated probability is claimed.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: These declared **weight**s are hand-chosen toy parameters; no network **training** or calibrated probability is claimed.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which connection explains why more right brightness lowers this **score**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The right connection has negative **weight** -1. Its **contribution** changes from -5 to-6 while the left **contribution** stays 6. Changed-case check: The **score** becomes 6-6=0.

The right **weight** is-1, so adding 1 to that **input** lowers the **score** by1.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Dot products and weighted sums: the case

02. Follow the deciding steps

03. Weighted brightness score

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.